

CSci 8314 Fall 2026

Sparse Matrix Computations

General Information

This course is an introduction to sparse matrix techniques and their applications. It will introduce sparse matrices in the context of linear systems of equations and discuss their origins and how they are stored and exploited. Then it will (very) briefly cover direct solution methods for solving sparse linear systems of equations. The remainder of the course will cover graph-based algorithms in machine learning, network analysis (centrality), graph embeddings, and related topics. It will also cover iterative methods for sparse linear systems and eigenvalue problems. One of the requirements of the course is the completion of a term project.

The course will consist of lectures given by the instructor and in class discussions. It qualifies as a 50% project toward a computer science plan C (coursework only) Masters.

- **Class Schedule:** 2:30pm - 3:45 pm M-W, Keller Hall 3-125
- **Instructor:** Yousef Saad <http://www.cs.umn.edu/~saad>
[e-mail: saad@umn.edu]
Office: 5 -225B Keller Hall – Office Phone: (612) 624 – 7804.
- **Prerequisite:** csci 5304 (or equivalent)
- **Office hours:** See class web-page

Textbook

There is not a single text book for the course but part of the lectures will be based on my book: *"Iterative methods for sparse linear systems (2nd edition)"* which is posted (free download) – see

<http://www.cs.umn.edu/~saad/books>.

This will be supplemented by articles and lecture notes.

Supplemental reading: *Direct methods for sparse linear systems*, by T. A. Davis, SIAM publishing, 2006. *You need not purchase the books listed above.*

Lecture Notes

Lecture notes on some of the chapters will be posted regularly, usually in advance of the lecture. These will be posted on github - see canvas for link. The lecture notes will be posted by topic rather than lecture by lecture.

Project

To pass this course, each student will need to complete a class project. This will generally be on the use of sparse matrix techniques in specific application areas. You can for example write a comprehensive survey on a selected topic related to the course. But you can also write a paper on how to exploit a specific class of sparse matrix methods in your own research. In the past the projects ranged from exhaustive survey articles on a specific topic (e.g., 'sparse matrix techniques in text mining') to implementations /comparisons of (a) specific method (s) say for solving sparse linear systems; or a specific theme in your own research that illustrates the use of sparse matrix methods. Minimal requirements, as well as the details on grading criteria, will be specified later in the semester.

Student Presentations: The current plan is to have each student who takes the class for credit give a 25mn presentation on their project. (Alternatively a few students will provide longer reports but will not present their projects). You can work alone or in a team of 2 [more if justified]. Since these presentations (and reports) constitute supplemental course material for the class, a good part of the grade will be based on their clarity.

Evaluation

The evaluation of your performance in this class will be based on the following:

- 65 % : *Term Project* - (based on report and presentation components)
- 21 % : *Three quizzes* at 7 % each [actually three best out of 4 quizzes given.] Details below
- 14 % : *Class participation and attendance*

Quizzes will be 10-15mn long each and will be related to the course material as well as the in-class exercises. There will be no make-up quizzes. The final letter grades will be assigned based on the following scale, where T is the total score (out of 100) you achieved in the class.

A : $100 \geq T \geq 92$	A- : $92 > T \geq 87$	B+ : $87 > T \geq 83$
B : $82 > T \geq 77$	B- : $77 > T \geq 72$	C+ : $72 > T \geq 65$
C : $65 > T \geq 60$	C- : $60 > T \geq 55$	D+ : $55 > T \geq 50$
D : $50 > T \geq 40$	F : $40 > T$	

Topics to be covered.

The topics to be covered will depend on participants and their interests and will be discussed and finalized during the first two lectures.

However these core topics will be covered: sparsity, working with sparse matrices, sparse linear systems of equations, applications to graphs, sparse graphs and graph Laplacian, graph partitioning, graph analysis (centrality measures), iterative methods for linear systems and eigenvalue problems.